

Theory Book

COMPUTER APPLICATIONS TECHNOLOGY

11



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA



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Computer Applications Technology Grade 11 Theory Book

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Dear learner

Welcome to the *Theory Book for Computer Applications Technology Grade 11*. Please read the information below about the book before you start using it. This book provides you with all the information you need to cover the Grade 11 Computer Applications Technology curriculum. It will also help you develop the skills, attitudes and values you need to be successful in your studies.

Each chapter in the book has the following features, which are meant to help you learn:

- **Overview**
This is an overview of the contents and the units of the chapter.
- **Learning outcomes**
These are the objectives that are addressed in the chapter. It states what you should be able to do by the end of the chapter and gives you an idea of what will be expected of you as a learner. After completing each chapter, ask yourself, 'Am I able to do everything stated in the learning outcomes?'. If not, you should revise the content covered in the chapter.
- **Introduction**
The introduction is a brief summary of why you will be learning the content that is covered in the chapter. Sometimes it will give you a brief background of how far technology has come, while at other times it will just give you a short summary that introduces the content that will follow in the chapter.
- **New words**
These are difficult words that you may not have encountered before.
- **Activities**
These cover questions you have to answer based on the content presented before them. Activities will help you check whether you have understood the content presented in the book or not.
- **'Something to know', 'Take note' and 'Tip' boxes**
The boxes provide extra, interesting content that might caution you to 'take note' of something important; or give you additional information. Note that the content in these boxes will not be part of your exams.
- **QR Codes, Videos and Screen captures**
These will link you to online content. When you are in the eBook, you can easily access the links.
- **Revision activities**
This is a revision activity based on what you have covered in the chapter. Take time to answer the questions on your own. Your teacher may also use these to assess your performance during class.
- **At the end of the chapter**
This is a checklist to ensure you understand all the content covered in the chapters. These criteria are linked to the learning outcomes at the beginning of the chapter.
- **Glossary**
Brief explanations for new words are given in the glossary at the back of the book.

Before getting started, watch the video in the QR code in the margin.



INTRODUCTION

Before getting started, watch the video in the QR code.



<https://www.youtube.com/watch?v=nKlu9yen5nc>

You can use the QR codes on these pages to link to online content for further information on these topics.

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GENERAL CONCEPTS OF COMPUTING

CHAPTER

1

CHAPTER OVERVIEW

Unit 1.1	Information processing cycle: input, processing, storage, output and communication
Unit 1.2	Computer features and classification
Unit 1.3	The role of ICTs in the workplace
Unit 1.4	The value of CAT

By the end of this chapter, you will be able to:

- Describe the information processing cycle
- Discuss different types of computers and their typical features
- Categorise computers
- Elaborate on the role of ICT in the workplace
- Discuss the value of CAT as a subject

INTRODUCTION

In Grade 10, you learned about the basics of computers. You were shown that all computers use the same basic process to function, namely the input, processing, storage, output and communication process. This is called the **information processing cycle**.

This chapter will refresh your knowledge on the basics of how computers work, the different types of computers there are and their typical features. You will then learn about how computers are categorised and how ICT has made the workplace faster and more efficient.

1.1 Information processing cycle

Before you can start to learn about the different types of computers that you will come across, you need to understand the basics of how computers work. All computers, whether they are the smartphone in your hand or large, powerful servers, operate on the same five basic principles. These are input, processing, storage, output and communication. Each component of a computer performs one of these functions, but they all work together to make the computer work.

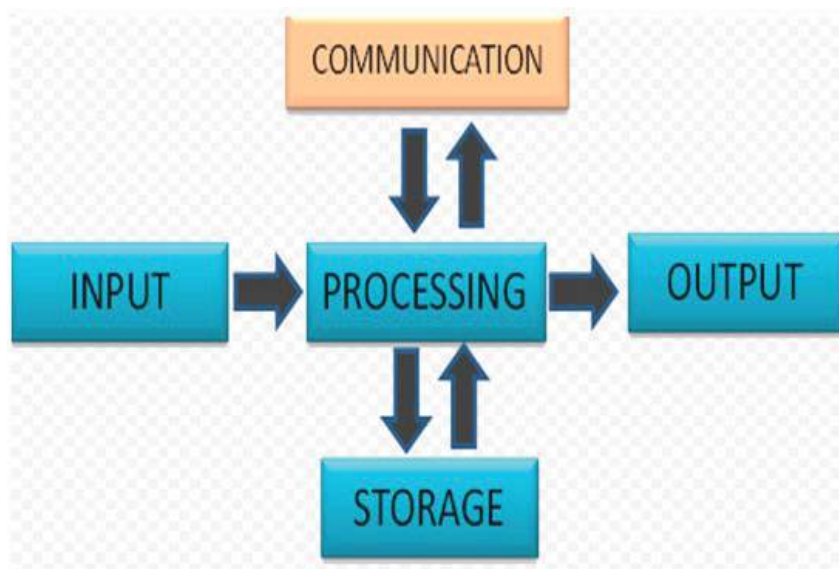


Figure 1.1: *The stages of the information processing cycle*

In this section we will look at each of these stages and how they work together. We will also look at how these processes can get one computer to communicate with users and other computers. The five main steps are input, processing, storage, output and communication.

INPUT

In the input stage, the data is entered into the computer. There are many ways to do this. In fact, there are as many ways to input data as there are input devices. You would have learned about input devices in Grade 10 but just to refresh your memory, input devices are things such as keyboards, touchscreens and microphones. The user inputs the data (for example, by typing on a keyboard or speaking into a microphone) into the computer. The device takes this data and converts it into a series of 1s and 0s (this is called **binary code**).

PROCESSING

The central processing unit (CPU) inside the computer then takes that binary code and does the calculations needed to get that data to display in a way that makes sense to the user. The CPU works with the computer's memory to get instructions on how to display the information from the input device and stores it as pixels in the computer's memory. This information is sent to the output device to be translated and displayed in a way that is useful. All of this takes a fraction of a second to do.

STORAGE

Storage is where the computer takes the input and stores it in its memory banks. There are many ways to store the data, but the basic process is as follows:

1. The CPU writes the data to the computer's temporary storage, or random access memory (RAM).
2. The computer then waits for the user's command to move the data from the RAM to more permanent storage. If that command is given, the computer writes the data to the disk drive.
3. Lastly, the computer saves the data in a location on the drive, either the default storage location or a location set by the user. The user can then recall this stored information at any time.

You can also store information using external storage devices (for example USB drives or external hard drives).

OUTPUT

Output is where the computer takes the pixels from the processing stage and displays them in a way that the user can see them. There are many kinds of output devices, such as printers, screens, video and audio devices.

These devices make the raw data usable and visible, allowing human users to interpret the data, turning it into information. This could be the sound waves of a song or the letters in a document.

COMMUNICATION

Communication is linked to the other stages in the information processing cycle. Each part of the cycle happens because one of the other steps came before it, meaning that they are linked. This link is the communication aspect of the information processing cycle. The most visible aspect of this is the relationship between input and output. Input and output happen almost immediately (for example typing a letter on a keyboard makes it appear almost immediately on a screen). This is communication between the user and the computer.

Communication can then further be sent across a network to other computers. Think about browsing the internet. The internet is basically a huge network of computers linked across the world. Opening your browser connects you to those other computers and these computers communicate with each other to give you what you are looking for. Computers that are linked in a network also communicate with each other. Examples of this are computers linked on a company's [intranet](#) or those linked to a server.

HOW COMPUTERS WORK



https://www.youtube.com/watch?v=OAx_6-wdSIIM

SENDING A WHATSAPP MESSAGE

To see how this works, think about sending a WhatsApp message from your smartphone. To start the process, you open WhatsApp and use the touch keyboard to enter your message (this is input, output and communication). Once you press SEND, your WhatsApp application communicates with the servers to send the message and displays it in the current conversation (this is communication, processing and output). The app keeps a record of the conversation on your phone, your friend's phone and the WhatsApp server (this is storage).

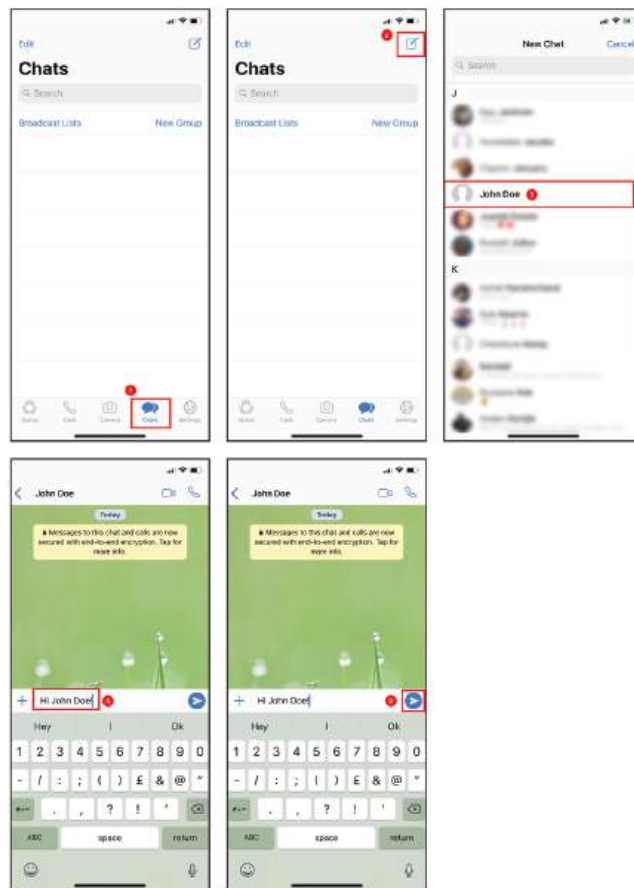


Figure 1.2: How sending a WhatsApp message works

As you can see, these processes are interlinked at every step of the way.



Activity 1.1

Do the following activity on your own.

1. Usually Mpumi uses the computers in her school's computer lab to do her homework. Mpumi's parents recently decided to get her a computer so that she can do more of her work at home.
 - a. What kind of computer should Mpumi use? Give two reasons for your answer.
 - b. List two types of input devices she would need.
 - c. List two types of output devices she would need.
 - d. List two types of storage devices she would need.
 - e. Mpumi wants to save a Microsoft Word document on her computer as "Geography.docx". Explain the basic process of how she will create and save this file on her computer. NOTE: Consider what happens during each stage of the information processing cycle.

1.2 Computer features and classification

Now that we understand how computers work, we can look at what types of computers there are and their typical components. Computers can be broadly separated into three categories, namely:

1. **Non-portable:** server, desktop
2. **Portable:** laptop, tablets, smartphones
3. **Dedicated devices:** ATM, smart fridge

These devices are classified according to how portable they are, what their processing power is and what they are used for.

This unit will look at these types of computers and their typical components, as well as look at how each type of computer can be classified.

COMPUTER CLASSIFICATION

Table 1.1 lists the different kinds of computers, what they are used for and their processing power.

Table 1.1: *Computing device classification*

DEVICE	USES	PROCESSING POWER	PORTABILITY	IMAGE
Servers	Their primary purpose is to serve information to users. They usually focus on one type of information.	They are designed to process massive amounts of data quickly, so have an extremely high processing power.	Not at all portable, since they are connected to each other in a network. Removing one means that it no longer functions as a server.	
Desktop PCs	General office and household use, including gaming, photo and video editing, budgeting and web browsing.	They can be extremely powerful, especially those designed for heavy use. In general, the processing power ranges from medium to very high.	Not very portable, since they are large and heavy. The internal components can be quite delicate and the PC will need an external power source.	
Laptops/ notebooks (portable devices)	Generally used for office work or personal use.	They are referred to as mid-range devices, meaning that they are suitable for normal use (document creation and editing, web browsing, etc.).	These devices are portable. They contain a battery, so do not always need an external power source. Most are lightweight with some, such as the Asus Zenbook, being extremely light.	
Tablets (portable devices)	Mainly used for entertainment or web browsing.	Their processing power is not very high.	Tablets are also portable as they are designed to be mobile. Tablets play an important role in the rise of mobile computing, which is the idea that you can take your computer anywhere.	
Smartphones (portable devices)	Designed mainly as a communication device and come with pre-installed messaging and voice-call apps.	They are not designed to be high-powered computing devices, although some do have quite a lot of processing power.	Smartphones are extremely portable since they are a blend of ultra-portable computing devices and mobile technology.	

DEDICATED DEVICES

Computers are everywhere around you, even in places that you least expect. It might surprise you to learn that computers are in **ATMs**, point-of-sale devices (that is, tills and card machines) and even fridges and air-conditioners.

These computers are called dedicated or embedded devices and their components will differ depending on what they are installed in, but they will most often consist of a display of some kind (either a screen or **light emitting diode (LED)** display) and some form of input device (such as a keypad or remote control).

They are usually limited in what they can and cannot do, as they are designed to perform very specific functions.

In recent years, so-called “smart devices” have become more popular and available. These devices are designed to interact with each other and the internet to make life easier for their users.

Table 1.2 lists different examples of dedicated and embedded devices.

Table 1.2: *Dedicated and embedded devices*

DEVICE	DESCRIPTION	IMAGE
ATM	ATMs are electronic banks that allow customers to draw and deposit money, pay bills and apply for loans without needing to visit a branch of the bank. ATMs are designed to only perform banking functions, so they are called dedicated devices. With the popularity of cryptocurrencies such as Bitcoin on the rise, there is now a new generation of ATMs that can handle cryptocurrency transactions. These Bitcoin ATMs can be used to buy and sell Bitcoin and other selected cryptocurrencies.	
Smart fridges	Smart fridges have touchscreens and are able to connect to the internet using Wi-Fi. Smart fridges can customise temperatures inside different compartments, track the expiry dates of food and alert you to when to use that food before it expires, and create grocery lists that sync with your smartphone. Different brands of smart fridges have different features. The computers inside smart fridges are called embedded devices.	



Activity 1.2

Do the following activity on your own.

1. Briefly describe what a dedicated device is.
2. Give five examples of a dedicated device.
3. Place the following devices in order of most portable to least portable.
 - A. Mainframe
 - B. Smartphone
 - C. Laptop computer
 - D. Desktop computer
4. Place the following devices in order of most processing power to least processing power.
 - A. Mainframe
 - B. Smartphone
 - C. Laptop computer
 - D. Desktop computer
 - E. Server
5. Determine which of the following computing devices are used in the following examples:
 - Desktop PC
 - Laptop
 - Tablet
 - Smartphone
 - Server

EXAMPLE	TYPE OF COMPUTING DEVICE
1. David works for a bank and needs to store petabytes of data on a computer system.	
2. Maseko wants to create a small graphic design company and needs 10 computers suited for this purpose.	
3. Mary-Anne travels a lot for business. During these business trips she needs a device she can use to create and edit documents.	
4. Darwin has a food blog on which he posts pictures of different locations where he has eaten. He needs to be able to take photos whenever he eats somewhere new.	
5. Carmen is very artistic and wants a computer that she can easily draw sketches on whenever she explores the countryside and beach.	
6. Emmanuel wants a simple and affordable computer that he and his family can use at home.	
7. Akshara is a college student who needs a computer she can practice programming on in-between lectures.	
8. Bongi needs a device she can use to socialise with her friends on Facebook, Twitter and WhatsApp.	
9. Minke has decided to launch a website so that she can sell her designer clothing online. This requires a computer that can store and manage information on her products, clients, orders, etc.	
10. Oliver is a game developer who works from home and needs a very powerful computer that he can easily upgrade over time.	

1.3 The role of ICTs in the workplace



COMPUTERS REPLACING HUMANS IN THE WORKPLACE?

ICT is changing the workplace
in South Africa:



[https://www.youtube.com/
watch?v=iULC9wHu1qc](https://www.youtube.com/watch?v=iULC9wHu1qc)

As you learned in Grade 10, **information and communications technologies (ICTs)** have an impact on every aspect of what people do at home, at school or at work. By looking at what you have learned so far in this chapter, you can see how ICTs have become a major component of daily life. This section will look at the role that ICTs play in the workplace.

The main reason ICTs are used in the workplace is that they allow humans to do their work faster, more efficiently and with fewer wasted resources. Companies no longer need staff to handle physical mail as email is faster and more effective than an employee sorting through post and delivering it to the correct people.

HOW ICTS IMPACT THE WORKPLACE

ICTs have also changed how workplaces are organised. ICTs allow employees to be more flexible in where, when and how they work, giving rise to the concept of mobile or virtual offices. Mobile offices are usually built for temporary purposes, usually within moveable, temporary buildings (such as an old shipping container). They can be completely virtual, with employees using mobile computing devices (such as laptops, tablets and smartphones) to create an office space outside of the business environment. Virtual offices can also be rented spaces that give businesses a physical address and office-related services (such as a telephone exchange) without the business needing to sign business leases or hire administration staff.



Figure 1.3: *ICTs connect staff working in different locations and aid communication between them*

Virtual offices are especially popular with new businesses, as there are lower starting and administrative costs, and a higher degree of flexibility and efficiency, since staff members do not have to handle day-to-day administrative tasks.

This has led to a change in how companies employ staff. The rise in flexi-time schedules (where staff work hours that suit them and not according to rigid timetables) and mobile offices has led to something called the decentralisation of labour. In the past, employees all worked at a central location according to a fixed time schedule (for example, at an office from 9 am to 5 pm). Now, employees can be scattered across the globe, all working at different times (that may or may not overlap with others) and in different places. Employers and employees are able to keep in touch using the internet and various software programs (such as Skype or email).

ICTs have also led to the rise of office automation. In some companies, ICTs have completely replaced the need for reception staff to answer telephones and take messages, due to messaging services and cellular technology.

OFFICES THEN AND NOW

A job that is slowly being phased out of existence due to ICTs is the receptionist or switchboard operator. Their job was to take calls and direct those calls to the correct department or person in a company. This is now handled by automated responses, like those used by large service providers such as MTN or Vodacom. When you dial their customer support number from your smartphone, an automated message plays, asking you to select a number to be put through to the correct department to help you with your query. In some cases, you might not even need to speak to a person, since there may be a computer on the other side of the line that is programmed to answer your questions with a series of automated responses.

Another change that ICTs have brought to the workplace has been the rise of **bring your own device (BYOD)** policies. As smartphones and tablets have become more cost-effective to buy, there has been a rise in the number of companies adopting a BYOD policy. BYOD means that businesses can avoid the costs of purchasing computing equipment for new staff members, as staff are encouraged to bring their own computers. Staff can also tailor their computers to their needs. However, BYOD is mostly only cost-effective for small to medium businesses, as larger businesses have the buying power to make arrangements with ICT providers to get the best devices for their staff.

BYOD policies also allow staff to take their devices home, meaning that businesses have a lower insurance risk as they do not need to keep expensive technology on their premises.



Activity 1.3

Do the following activity on your own.

1. As technology is always changing and improving, people are finding more ways to use technology to make certain processes in the workplace more efficient.
 - a. What are ICTs and why are they used in the workplace?
 - b. In the workplace, how are ICTs used to improve the way employees are paid?
 - c. In the workplace, how are ICTs used to improve the way employees communicate with each other? Also mention TWO software programs that employees can communicate with.
2. Give TWO examples of how ICTs can be used to improve a school environment.
3. Name TWO jobs, which currently exist, that you think will be replaced by ICTs in the future. Give a reason for each of your answers.



WANT A JOB IN 2020?



<https://www.youtube.com/watch?v=Y9F0yoS3Fag>

1.4 The value of CAT

We are living in the information age so it is important for people to understand the value of computers and how to use them. This is where CAT as a subject comes in. By understanding computers and their basic components, programs and applications, you will be able to adapt to new developments in computing quickly, easily and with less training. CAT teaches you the basic skills you will need to prosper in any career, even those that use specialised software and equipment.

COMPUTER LITERACY AND EMPLOYMENT

Apart from using your computer to finish your schoolwork or do research for projects, knowing how to use a computer is a requirement for almost any job today. From obvious jobs, such as computer programmers and graphic designers, to less obvious jobs, such as truck drivers and carpenters, each person entering the job market must know how to use a computer with some degree of skill.

With computers and the internet becoming more accessible, an entirely new workspace has opened up, where digitally savvy people are using computers and computing technology to advertise, promote and build businesses online. Online marketing (or digital marketing) is one of the fastest growing job sectors in the world and skilled digital marketers are in high demand.

It is not only people who will be working in ICT-related fields who will need to be computer literate. In 2017, a News Corp Australia article found that by 2020, about 90% of the Australian workforce would need to have basic computer skills to communicate with others and to find information about their tasks and daily duties.

CAT is also an incredibly powerful tool when it comes to managing information. By knowing how you can use a computer to manage large amounts of data and information, you will be able to work quickly and more efficiently in your future career. CAT teaches you how to use programs and systems to manage data and information.

CAREER OPTIONS AND THE IMPACT ON CAREERS AND FIELDS OF STUDY

Taking CAT at school level opens the door to many possibilities and career options once you have completed your schooling. Almost all careers available to you use computers in some part of their daily operations. From fashion, web and graphic design to accounting, law and medicine, having CAT as a subject will allow you to use the technologies associated with these careers. A good understanding of computer applications technologies also opens up Information Technology and Management to you as a field of study and work.

Studying CAT will also give you insight into how the internet and the World Wide Web function and how they can be used to enhance your knowledge and life.



Activity 1.4

Do the following activity on your own.

1. Bernes is a learner who dreams of becoming a doctor when he grows up. Bernes has decided that he does not need to take CAT because it is not relevant to his dream job.
 - a. What job would you like to do in the future? Mention three ways in which CAT can prepare you for this job.
 - b. Name three ways in which CAT could improve Bernes' study life as a university medical student.
 - c. Do you think that people in the medical field need to know how to work with computers? Give a reason for your answer.
 - d. While taking a tour of one of the best medical universities, Bernes discovers that a lot of the medical equipment makes use of computers.
 - i. What type of computer is used in medical equipment such as heart rate monitors?
 - ii. Give two reasons why you think a lot of medical equipment makes use of computers.
2. Do you know of a job that cannot benefit from CAT? If yes, mention it and give a reason for your answer. If no, explain your answer.

REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

- 1.1 Which of the following is not part of the information processing cycle? (1)
 - A. Communication
 - B. Output
 - C. Storage
 - D. Memory
- 1.2 Which of the following can be used for storage? (1)
 - A. CPU
 - B. HDD
 - C. GPU
 - D. ICT
- 1.3 Which of the following computers is not portable? (1)
 - A. Laptop
 - B. Tablet
 - C. Smartphone
 - D. Server
- 1.4 Which of the following computers can be used to process large amounts of data quickly? (1)
 - A. Mainframe
 - B. Workstation
 - C. Tablet
 - D. Notebook
- 1.5 Which of the following is not a "smart device"? (1)
 - A. ATM
 - B. Smart light bulb
 - C. Calculator
 - D. Xbox 360

... continued

REVISION ACTIVITY

QUESTION 2: TRUE OR FALSE?

Choose the answer and write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)

- a. Taking CAT at university level opens up many career possibilities. (1)
- b. New businesses can lower their administrative costs by using a centralised office. (1)
- c. Printed statements have been mostly replaced by electronic statements sent via email. (1)
- d. CAT as a subject teaches us how to use computers so that we can become digitally savvy. (1)
- e. Networking is part of the communication stage. (1)

QUESTION 3: MATCHING ITEMS

Choose a concept from Column B that matches a description in Column A. Write only the letter next to the question number (e.g. 1J). (5)

COLUMN A	COLUMN B
1. A device that connects a computer to other computers in a network.	A. Laptop
2. A device with medium processing power that uses miniaturised parts.	B. Smart fridge
3. A device that stores large amounts of data and processes data at extremely high processing speeds.	C. Smart light bulb
4. A device that reduces the amount of energy that a household consumes.	D. Smartphone
5. An extremely portable device that mainly makes use of mobile technologies.	E. ATM
	F. CPU
	G. Server
	H. NIC

QUESTION 4: SHORT AND MEDIUM QUESTIONS

- 4.1 Briefly explain what happens during the following stages of the information processing cycle.
 - a. Processing stage (2)
 - b. Storage stage (2)
- 4.2 Compare the functions of input and output devices. (4)
- 4.3 Name a device that can be used to electronically transfer money. (1)
- 4.4 What is the relationship between CAT and ICTs? (2)
- 4.5 Give the name of a non-portable device that is suitable for creating documents and web browsing. (1)

... continued

REVISION ACTIVITY

QUESTION 5: SCENARIO-BASED QUESTIONS

Adam and his family have decided to open a sweet shop called the Sweet Company. As his shop starts to grow, he decides to open a second Sweet Company at a different location. In order to keep track of both shops, he decides to start integrating ICTs and computers into his business.

- a. Name TWO computing devices Adam and his employees can use to better run and manage his shops. Give a reason for each of your answers. (4)
- b. Mention THREE ways in which ICTs can improve the way that Adam's sweet shop is run. (3)
- c. Mention TWO instances where CAT skills can be used to improve the running of Adam's sweet shops. (2)
- d. Give TWO reasons why Adam should consider purchasing a UPS for both shops. (2)
- e. Adam can now work on a flexi-time schedule. Explain what it means to work flexi time. (2)

TOTAL: [40]

AT THE END OF THE CHAPTER

NO	CAN YOU ...	YES	NO
1.	Describe the information processing cycle and how it relates to how computers work?		
2.	Distinguish between the different types of computers in terms of their processing power, uses and portability?		
3.	Describe how to categorise computers?		
4.	Define the role of ICTs in the workplace?		
5.	Discuss the value of CAT as a subject and how CAT can influence your future career?		

CHAPTER
2INPUT, OUTPUT AND
OTHER DEVICESCHAPTER OVERVIEW

Unit 2.1 Input devices

Unit 2.2 Output devices

Unit 2.3 Devices, software and equipment

By the end of this chapter, you will be able to: 

- Describe various types of input devices, including scanners, digital cameras, biometric input devices and others
- Discuss the advantages, disadvantages and limitations of various input devices
- Explain how the quality of scanners and digital cameras is measured
- Describe various types of output devices, including display devices and interactive whiteboards
- Discuss the advantages, disadvantages and limitations of various output devices
- Explain what criteria are used to measure the quality of display devices
- List and describe the software and equipment needed to operate input and output devices
- Discuss the purpose of optical character recognition (OCR) and hand-held OCR devices

INTRODUCTION

In Grade 10, you learned that hardware is the physical components of a computer. This includes items such as the monitor (or screen), keyboard and mouse.

An input device is any device that allows you to enter data into a computer and interact with it. Common input devices include keyboards, computer mice, touchpads and touchscreens. You also learned about the basics of digital cameras, scanners and readers such as radio-frequency identification (RFID), magnetic strip and OCR readers. Other input devices are video and audio input devices such as webcams and microphones, and biometric input devices such as fingerprint scanners.

Output devices take the processed input from a computer and display it in a way that is easy for humans to understand. Screens are the main output devices of any computer. Liquid crystal displays (LCDs) and LED screens are the most popular types. Printers are another common type of output device. There are two main printer types, namely inkjet and laser printers.

Headsets and speakers are designed for audio output, with other output devices being fax machines, multifunction devices (which combine faxing, emailing and printing) and data projectors.

Processing components include hardware such as the:

- Motherboard, which connects the components in a computer and houses the ports, such as the universal serial bus (USB), video graphics array (VGA) and **high-definition multimedia interface (HDMI)** ports to connect input and output devices.
- Central processing unit (CPU), which receives and carries out the instructions inputted by the user.
- Graphics processing unit (GPU), which makes the calculations and follows the instructions necessary to display images on a screen.

Storage devices are the computer components designed to keep (or store) data. This data can be the information needed to make the computer function, such as the operating system or **basic input/output system (BIOS)**, or data created by the user, such as images, documents, text files and so on.

These components, called storage media or storage devices, are any piece of computing hardware used to keep or store data files. They can hold and store information permanently or temporarily and can be internal or external.

Internal storage media, such as hard drives and RAM, are inside a computer and part of it, while external hard drives and USB drives are outside a computer and can be removed easily and quickly.

2.1 Input devices

As you learned in Chapter 1, a computer works using the information processing cycle. Input devices are the key components of the first stage of the cycle, the input stage. Input devices are what we use to interact with a computer. These can be things such as keyboards and computer mice, touchpads and scanners. The combination of keyboard and mouse used to be the most common input device, but the rise of the smartphone has made the touchscreen the most popular and common input device in the modern age.

There has also been a rise in the use of alternative input devices, such as fingerprint and face recognition to unlock your smartphone, and speech-to-type devices that are used by people with physical challenges.

This unit focuses on input devices other than the mouse and the keyboard as they were covered in Grade 10.

VARIOUS INPUT DEVICES

There are a number of input devices that you can use with computers. Table 2.1 lists these devices, their uses, and their advantages and disadvantages.

Table 2.1: *Input devices*

NAME AND IMAGE	PURPOSE	WHEN TO USE	ADVANTAGES	DISADVANTAGES
Scanners 	Convert physical data (documents or photographs) into electronic data.	Scanners are used whenever you need a digital copy of a physical (hard copy) document.	- Produce accurate and high-resolution images. - Scanned images can be added to electronic documents or edited easily.	- Scanned images can lose clarity and quality. - Image files can be very large and take up space. - The quality of the original affects the quality of the scanned image.
Digital cameras 	Digital cameras are used to capture digital images and input them directly into a computing device.	- Digital cameras can be used to capture images. - The digital cameras found in smartphones can also be used to record video.	- You can view your images immediately and discard any that you do not want. - Digital cameras are more environmentally friendly because you do not need to use toxic chemicals to develop the film.	- The quality of the image will be limited to the maximum resolution of the digital camera. - Digital cameras have certain limitations for nighttime photography.
Biometric input devices 	Biometric input devices use the unique measurements and characteristics of your body to interact with a computing device.	- Fingerprint and face scanners in smartphones. - Eye and iris scanners. - Voice recognition software.	- Faster to use than passwords. - Biometric input is harder to hack.	- It is influenced by disease and substances. - Some people's fingerprints cannot be read. - Biometrics are for life and cannot be changed. - There are some privacy concerns about collecting biometric data.

... continued

NAME AND IMAGE	PURPOSE	WHEN TO USE	ADVANTAGES	DISADVANTAGES
ATMs and point-of-sale (POS) terminals 	ATMs and POS terminals are specialised input devices.	- ATMs are used in the banking industry to deposit or withdraw cash and get statements. - POS terminals are used in the retail and restaurant industries to track stock and orders.	- ATMs make basic banking faster and easier. These features are available 24/7. - POS terminals can keep accurate track of stock and can provide reports on sales and stock movement.	- ATMs can be targets of card-skimming scams and using ATMs to draw cash can make you a target for criminals. - POS systems also need to have their software updated and staff must be trained to use them.
Wireless input devices 	These are usually keyboards and computer mice that do not use cables to connect to a computer.	Wireless devices can be used with almost any computing device.	Wireless devices are not limited by cables and can be used anywhere.	- Distance can affect the functionality of these devices; issues may be experienced the further away they are. - They have limited by battery life.
Handheld devices 	These devices include smartphones and tablets.	Handheld devices such as smartphones and tablets can be used to make calls and send instant messages. They can also be used if you do not have access to a traditional computer.	Handheld devices can connect you to the world.	- Their performance is limited by battery life. - Potential for information to be leaked. - Increased risk of damage to device.
Touchscreens 	Touchscreens are used as the primary input devices for smartphones and tablets. They are also used on laptops and PCs, ATMs, POS terminals and in biometrics.	The most common use for touchscreens is as the input method for smartphones and tablets.	Touchscreens are more versatile and compact.	- Touchscreens have to be kept clean to function optimally. - Limited by battery life. - Increased risk of damage.
Alternative input devices 	These devices are used as alternative input methods.	The most common use for alternative input devices is the virtual keyboard.	Virtual keyboards are extremely reliable.	It does not offer tactile feedback.

WHAT DETERMINES THE QUALITY OF THE IMAGE TAKEN BY A SCANNER OR CAMERA?

There are three main factors that determine the quality of the image taken by a scanner. These are:

1. Colour depth
2. Resolution
3. Dynamic range

Colour depth is also known as **bit** depth and refers to the number of bits used to indicate the colours of a single pixel. The higher the bit number, the better the colour depth. You can see this in Figure 2.1.



Figure 2.1: 32-bit (left) versus 8-bit (right) colour depth.

The image on the left is in 32-bit colour while the image on the right is in 8-bit colour. In the image on the left, the details in the background are sharper and the colour of the leaf is deeper and more vibrant compared to the image on the right.

Resolution is the amount of detail an image can hold and it is measured in **pixels per inch (ppi)** or **dots per inch (dpi)**. These measurements show you how many dots or pixels are in a one-inch square (an inch is about 2,5 cm). The higher the ppi or dpi, the more information there is in the square. This means that the image will be of higher quality.

The final quality factor is the *dynamic range*. This measures the range of light the scanner can read and use to produce a range of **tones** and colours.

Camera quality is determined by three factors:

1. Resolution
2. Lens aperture
3. Focal length

Resolution is the amount of detail that a camera can capture. In digital cameras, resolution is measured in **megapixels**.

The *lens aperture* is the maximum amount that the lens can open. The wider it opens, the more light it can take in, which means that you need less light to take a good picture.

How much a camera can zoom is determined by its *focal length*. The focal length is shown by a number and the times symbol (×). A zoom of 3× means that the longest focal length is 3× the distance of the shortest focal length.



Activity 2.1

1. Multiple choice:
 - a. Which of the following does NOT determine the quality of a digital camera?
 - A. Resolution
 - B. Focal length
 - C. Price
 - D. Lens aperture
 - b. Which of the following is NOT an example of biometric input devices?
 - A. Voice recognition
 - B. DNA tester
 - C. Fingerprint scanner
 - D. Facial recognition
 - c. Which device CANNOT be used when you need to convert a physical copy of a document into a digital copy?
 - A. Digital camera
 - B. Scanner
 - C. Smartphone
 - D. Laser keyboard

2. Match Column B with Column A:

COLUMN A	COLUMN B
2.1. The amount of detail that a camera can capture, measured in megapixels.	A. Pixels per inch
2.2. The category of devices used to interact with a computer.	B. Bluetooth keyboard
2.3. Another name for a wireless keyboard.	C. Resolution
2.4. Devices that link to video-calling applications using the internet.	D. Lens aperture
2.5. Used in the retail and restaurant industries to track stock and orders.	E. Touchscreens
2.6. The primary input devices for smartphones and tablets.	F. Webcams
2.7. The maximum amount a camera lens can open.	G. Input devices
2.8. The measurement for scanner resolution.	H. Point-of-sale terminals

3. Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)
 - a. It is good practice to regularly back up your files.
 - b. A POS system does not need to be maintained after installation.
 - c. Touchscreens can be used for biometric scanning.
 - d. Resolution is the amount of colour an image can show.
 - e. Dynamic range measures the range of light a scanner can read.

... continued



Activity 2.1

... continued

4. Answer the following questions in your own words:
 - a. What factor determines how much a camera can zoom?
 - b. Give three advantages and three disadvantages of scanners as input devices.
 - c. Briefly describe the factors that determine a scanner's quality.
 - d. Why would a Nikon Coolpix B500 or a Canon 4000D camera not be considered an input device?
 - e. Describe why digital cameras in smartphones become outdated very quickly.
 - f. Describe the factors that determine a camera's quality.
 - g. How would you use fingerprint scanners at a school?
 - h. Describe why a criminal might clone your fingerprints.
 - i. What are the advantages of POS systems and ATMs?
 - j. Why are touchscreens becoming more common?
 - k. Give two types of wireless input devices.
 - l. Give one disadvantage of laser keyboards.

2.2 Output devices

Output devices form the last stage of the information processing cycle. These are devices that translate the information from the internal processing components of the computer and display them in a way that makes sense to a human user. This could be text, images, sounds or videos.

There are as many output devices as there are input devices and there will certainly be more in the future as the way we interact with computers changes. When you think about output devices, you no doubt think about screens and displays, printers (including 3D printers), speakers and headphones. But interactive whiteboards, data projectors and GPS devices are also output devices.

In this unit you will focus on interactive whiteboards and displays and the advantages, disadvantages and limitations of each of them. You will also learn about the factors determining the quality of display devices and printers, as well as the wireless technology needed for these devices to operate correctly.

DISPLAY DEVICES

Display devices are a broad category of output devices that includes monitors (computer screens), smartphone screens and data projectors.

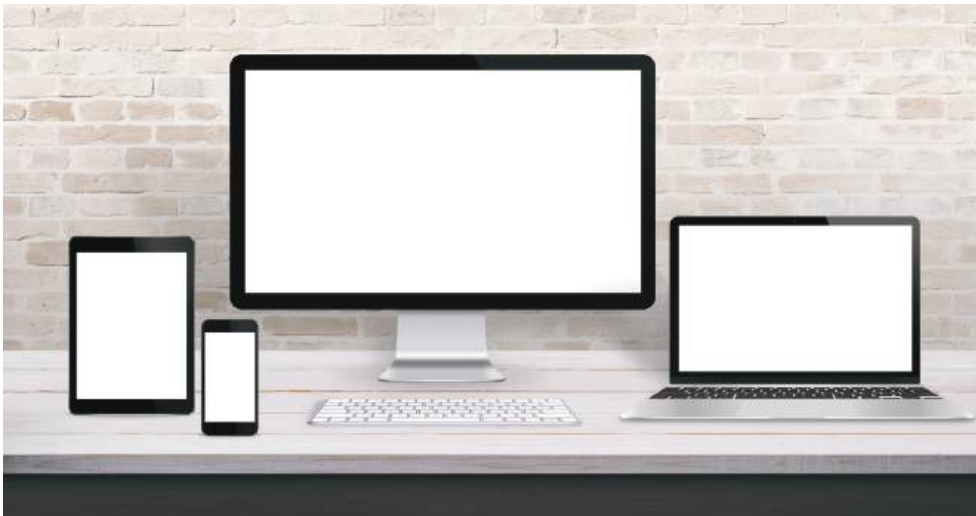


Figure 2.2: Examples of display devices

The main function of a display device is to display the input you have given the computer to allow you to make choices and interact with programs and the operating system.

There are some display devices that act as both input and output devices; touchscreens and interactive whiteboards are examples of these types of devices.

Display devices have several advantages, for example:

- The user gets immediate feedback about what is being processed.
- The interaction between the user and the computer is easier.

- Content can be displayed and changed quickly (think of switching between applications on a smartphone or programs on a computer).
- Displays can be customised to suit the user (for example, making the text size larger or lowering the brightness to make things easier to read).

Display devices also have several key disadvantages:

- The content that is displayed is not private. Anyone behind you can see what you are doing on your computer. This is less of a problem with smartphones, as the screens are easier to hide.
- Displayed output is not permanent and you would need to print or save the data to have a permanent record of what you see.
- Display devices are the one component that consumes the most power. Screens are usually the biggest drain on a smartphone, tablet or laptop's battery.
- They are usually the most expensive item to replace when they break.

WHAT DETERMINES A DISPLAY DEVICE'S QUALITY?

Several factors determine a display device's quality. These include resolution, aspect and contrast ratios and response time, as well as other factors. Less expensive devices usually have very high specifications in one area and low specifications in another. Devices that have a good balance of specifications are usually more expensive.

Table 2.2 lists the quality specifications you should look for when buying a new display device and gives a brief explanation of each specification.

Table 2.2: *Quality specifications for display devices*

SPECIFICATION	DESCRIPTION
Resolution	In the case of display devices, resolution is the number of pixels that can be displayed horizontally and vertically. For example, a monitor with a resolution of 1 920 x 1 440 can display 1 920 pixels horizontally and 1 440 pixels vertically.
Aspect ratio	This is the basic shape of the screen based on the ratio of the width to the height measured in inches. For example, if a screen is 16 inches (40,6 cm) wide and 10 inches (25,4 cm) tall, the aspect ratio is 16:10.
Contrast ratio or colour depth	This is a measure of the number of shades the display device can show between its blackest black and brightest white. The higher the number of shades, the clearer and sharper the images will be. This will also mean that the colours will be brighter and truer to life.
Response time or frequency	Response time measures how quickly the pixels on a screen refresh. Faster response times mean that fast-moving scenes on a video will be smoother and less blurry. This is especially important when the screen is used to watch movies or play games.

PRINTER QUALITY

There are many factors that can affect the quality of the prints that come out of a printer. The most important factor is the resolution, which is measured in dots per inch (dpi). As you learned in the section on scanners, dpi measures how many dots there are in a square inch of an image and the higher the dpi is, the better the image quality will be.

In printers, dpi measures how well the printer can match the pixels per inch (ppi) of a digital image or text. If the printer can match the ppi exactly, the image will come out crisp and clear and will look the same when it is printed and when it is on screen. The problem comes in when the printer cannot match the ppi, so the image quality goes down.

How quickly a printer can print is another indicator of the quality of a printer. This is measured in pages per minute (or ppm) and the higher the ppm, the faster the printer prints. Laser printers will always have a higher ppm than inkjet printers because of the different ways that they print.

Lastly, the duty cycle also shows the printer's quality. The monthly duty cycle indicates the maximum number of pages a printer can print in a month without it failing. A typical home printer will have a duty cycle of around 5 000 pages, while large business printers will have a duty cycle of around 200 000 pages per month.

INTERACTIVE WHITEBOARDS

At first glance, interactive whiteboards look exactly the same as normal whiteboards. But looking closer reveals that they are unique and an exciting tool to use in the classroom and in business. Interactive whiteboards can be seen as input and output devices.

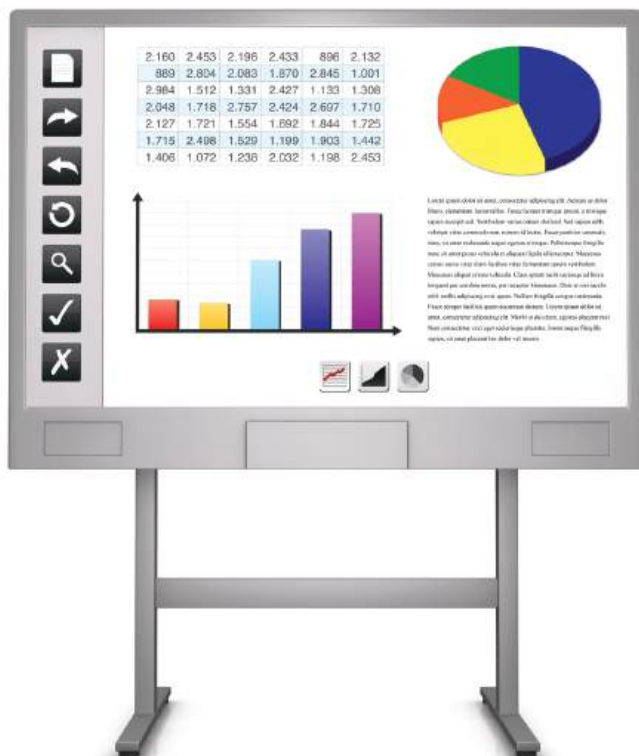


Figure 2.3: Example of an interactive whiteboard



Something to know

A printer's dpi is usually shown just like a screen's resolution is shown, since it is a measure of how many dots can be made horizontally and vertically. For example, a printer's dpi may be shown as 1 600 × 2 000.



HOW PRINTERS WORK

Ink printer:



<https://www.youtube.com/watch?v=9yeZSaigBj4&t=10s>

Laser printer:



<https://www.youtube.com/watch?v=WB0HnXcW8qQ>



Something to know

A normal non-magnetic whiteboard (2 000 mm × 1 200 mm) cost about R1 750 in 2018. This is one of the biggest whiteboards available.

Interactive whiteboards can be used like traditional whiteboards but you need to use special pens (that contain no ink) to write or draw on them. What makes interactive whiteboards so special is that they can be linked directly to a computer and can display what is on the computer. The clue to what sets these apart from normal data projectors is given in the name.

Interactive whiteboards allow users to interact with the data that is displayed. By using a mouse or even your finger, you can drag, click and copy items. You can use your finger or the special pens to write notes, highlight sections or make comments. Anything that is done on the whiteboard can then be saved and shared.

Interactive whiteboards are an incredible tool for teaching and have three key advantages when used in a classroom:

1. Interactive whiteboards allow teachers to combine visual and **auditory** learning styles into one. This allows learners to absorb information in multiple formats.
2. Because the notes made on the whiteboard can be saved and shared, learners do not need to take notes during a lesson. This allows them to focus more on participation during the lesson.
3. Interactive whiteboards allow learners to become part of the learning process by making them active participants in their lessons.

Some of the notable disadvantages of interactive whiteboards include the cost, training and time requirements. Most interactive whiteboards are incredibly expensive (in 2018, an interactive whiteboard cost about R6 100) and they need special equipment to work correctly.

Teachers and other users also need to be trained in how to use interactive whiteboards correctly and teachers may have to spend extra time preparing proper interactive lessons to get the most out of them.

WIRELESS TECHNOLOGY

Wireless output devices use Wi-Fi, **Bluetooth** or **near-field communication (NFC)** to communicate with the computer. Wireless devices reduce the number of cables you need to connect them to each other. Some examples of wireless output devices are wireless printers and wireless headphones or speakers.

Wireless printers can be connected to multiple devices on the same wireless network, meaning that all the people in a specific area have access to the printer. Wireless printers can also connect to smartphones and tablets, so users can connect to them directly. However, since anyone can access a wireless printer, there are some safety and privacy concerns surrounding them.

Wireless audio devices, such as speakers and headphones, use Bluetooth to connect to the computer. These output devices give users more freedom of movement as they are not physically attached to their computers.



Activity 2.2

1. Multiple choice:
 - a. Which of the following cannot be a wireless output device?
 - A. Monitors
 - B. Printers
 - C. Headphones
 - D. Speakers
 - b. Which of the following is NOT a wireless communication medium?
 - A. Wi-Fi
 - B. Bluetooth
 - C. NFC
 - D. RSS
 - c. Which of the following is the duty cycle of a home printer?
 - A. Around 500 pages
 - B. Around 5 000 pages
 - C. Around 10 000 pages
 - D. Around 1 000 pages
2. Match Column B with Column A:

COLUMN A	COLUMN B
2.1 An indication of a printer's resolution.	A. Touchscreen
2.2 The basic shape of the screen based on the ratio of the width to the height measured in inches.	B. Interactive whiteboard
2.3 A device that is both an input and output device.	C. Contrast ratio
2.4 A measure of the number of shades the display device can show between its blackest black and brightest white.	D. dpi
2.5 You need to use special pens (that contain no ink) to write or draw on this device.	E. Aspect ratio

3. Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)
 - a. Most wireless audio devices, such as speakers and headphones, use Bluetooth to connect to the computer.
 - b. A printer's dpi can be shown in the same way a screen's resolution is shown.
 - c. Interactive whiteboards allow teachers to combine different visual learning styles into one.
 - d. Interactive whiteboards are the same as traditional whiteboards.
4. Answer the following questions in your own words:
 - a. What would the aspect ratio be of a monitor that is 20 inches wide and 15 inches tall?
 - b. Name two extra features that determine the quality of a display device.
 - c. What are the three key advantages of interactive whiteboards in the classroom?
 - d. Explain why every school in South Africa does not have interactive whiteboards.
 - e. Give two advantages of wireless printers.

2.3 Devices, software and equipment

Often you will need additional software or hardware to make sure that your input and output devices can communicate with your computer. In most cases, your hardware will require a device driver to make sure that the commands it sends to and receives from your computer will be interpreted correctly.

In this unit, you will learn more about device drivers that you need to install yourself and **plug-and-play devices**. You will also look at the physical ways devices connect to your computer. Lastly, you will look at what OCR is and where it is used.

DEVICE DRIVERS

Device drivers are software programs that are separate from the device. They are either given on a disk or downloaded from the manufacturers' websites. Your OS will use a device's driver to communicate with the device. The driver will help the OS understand the information you give it through an input device or translate the information the computer sends to an output device so that it can display the information correctly.

Most operating systems also have a set of standard device drivers installed for things such as computer mice, keyboards and hard drives (internal and external). These devices are often referred to as plug-and-play devices.

PLUG-AND-PLAY

Plug-and-play devices are usually external devices but can also be some internal components of the computer (such as graphics cards). Examples of plug-and-play devices are headphones and speakers, computer mice, keyboards and USB storage devices.

OTHER EQUIPMENT

Most input and output devices need to be connected to the computer in some way for them to work. Keyboards and computer mice need to be connected either with a wireless USB dongle or a wired USB connector. Monitors will need power cables as well as display connectors (such as HDMI or VGA cables). Printers and projectors can be connected to the computer with cables, USB dongles or Bluetooth connections.

OCR

OCR stands for optical character recognition and it is a software program designed to convert printed characters (such as those on a hard copy document) into digital text. This means that you can edit and search in a scanned document in a word-processing program.

OCR is most commonly found in specialised scanners used for archiving or document management. You can also get handheld OCR scanners that allow you to quickly scan printed text into a digital format. OCR enhances traditional scanning, since it saves the documents as text and not as an image.

OCR can also be used to improve accessibility for physically challenged users. OCR scanners can work alongside text-to-speech devices to read printed text aloud. OCR scanners are also used in handheld barcode scanners.



Activity 2.3

1. Multiple choice:
 - a. Which of the following is not a plug-and-play device?
 - A. Speakers
 - B. Keyboard
 - C. Motherboard
 - D. Headphones
 - b. Which of the following is not a connection method for a monitor?
 - A. HDMI
 - B. Bluetooth
 - C. VGA
 - D. Power cable
2. Match Column B with Column A:

COLUMN A	COLUMN B
2.1. A software program designed to convert printed characters into digital text.	A. Device driver
2.2. The software that allows the hardware to work.	B. Monitors
2.3. Devices that can be connected to a computer and used immediately.	C. OCR
2.4. These devices will need power cables and display connections to work correctly.	D. USB
2.5. A type of connection for wired input and output devices.	E. Plug-and-play

3. Answer the following questions in your own words:
 - a. Where can you get device drivers?
 - b. Explain how device drivers work.
 - c. How can OCR be used to improve accessibility for physically challenged users?

REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

- 1.1 Which of the following is a disadvantage of biometrics? (1)
- A. It cannot be changed.
 - B. It is expensive to implement.
 - C. It is easy to use and requires little training.
 - D. It is difficult to lose.
- 1.2 Colour depth is _____. (1)
- A. The number of bits used to indicate the colours of a pixel.
 - B. The amount of detail an image can hold.
 - C. The range of light that can be read and used to produce a range of tones and colours.
 - D. The number of pixels used in a 1-inch square.
- 1.3 Which of the following does not use device drivers? (1)
- A. Mouse
 - B. Touchscreen
 - C. Windows Defender
 - D. Printer
- 1.4 Which of the following is NOT usually a plug-and-play device? (1)
- A. Mouse
 - B. Speaker
 - C. Keyboard
 - D. Graphics card
- 1.5 The signals from wireless devices cannot be interrupted by _____. (1)
- A. Bluetooth
 - B. Radio waves
 - C. Microwaves
 - D. Fridges

QUESTION 2: TRUE OR FALSE

Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)

- a. Voice recognition can be used to help people with hearing disabilities. (1)
- b. Fingerprint scanners use biometrics to identify people. (1)
- c. Scanners use OCR to convert digital text to printed text. (1)
- d. Printed images can become corrupted so you will need to back them up regularly. (1)
- e. A POS system is not an example of an output device. (1)

... continued

REVISION ACTIVITY

... continued

QUESTION 3: MATCHING ITEMS

Choose a term/concept from Column B that matches a description in Column A. Write only the letter next to the question number (e.g. 1–J).

(5)

COLUMN A	COLUMN B
1. An input device that can only be used in dim light.	A. Aspect ratio
2. Devices that use the unique measurements and characteristics of your body to interact with a computing device.	B. Scanner
3. The ratio of a screen's width and height in inches.	C. Focal length
4. The refresh rate of the pixels on a screen.	D. Resolution
5. A device that converts digital data to printed data.	E. Frequency
	F. Display device
	G. Laser keyboard
	H. Printer
	I. Biometrics

QUESTION 4: SHORT AND MEDIUM QUESTIONS

4.1 Look at the following image and answer the questions that follow.



- What type of input device does the image show? (1)
 - Name two other types of input devices that make use this type of device. (2)
 - Mention one disadvantage of this device. (1)
 - Give two reasons why this input method is the most natural way of interacting with a computer. (2)
- 4.2 Provide two advantages of display devices. (2)
- 4.3 Mention two jobs that can benefit from using interactive whiteboards. Give an explanation for each of your answers. (4)

... continued

REVISION ACTIVITY

... continued

QUESTION 5: SCENARIO-BASED QUESTIONS

Thomas is a high-school teacher who teaches a small class of 15 learners. Some of Thomas's learners have physical challenges that would make it difficult for them to interact with a traditional computer. Thomas therefore has to think of what types of extra input and output devices he will need to make this class project work.

- 5.1 Name two input devices that Thomas can use for a learner who cannot use their hands. (2)
- 5.2 List two output devices that Thomas's learners can use for their class projects. Also mention what equipment Thomas would need to connect each of these devices to a computer. (4)
- 5.3 Explain one way in which physically challenged learners can benefit from OCR technology. Also mention what type of input and output device will be used. NOTE: Mention the benefit and the related physical challenge. (4)
- 5.4 Thomas needs to order a printer. He wants to use it to print out pictures of his class and the best class projects.
 - a. What kind of printer should he order? (1)
 - b. What specifications should he look for to get good quality prints. (2)

TOTAL: [40]

AT THE END OF THE CHAPTER

NO	CAN YOU ...	YES	NO
1.	Describe various types of input devices, including scanners, digital cameras, biometric input devices and others?		
2.	Discuss the advantages, disadvantages and limitations of various input devices?		
3.	Explain how the quality of scanners and digital cameras is measured?		
4.	Describe various types of output devices, including display devices and interactive whiteboards?		
5.	Discuss the advantages, disadvantages and limitations of various output devices?		
6.	Explain what criteria are used to measure the quality of display devices?		
7.	List and describe the software and equipment needed to operate input and output devices?		
8.	Discuss the purpose of OCR and hand-held OCR devices?		